

General notes for all projects

- Each project can be selected solely by a single group. Formation of groups is upon your consent that must be **registered** in Moodle's **Choice** activity. The group size is **6** students; the total number of groups is 12. As there will be exactly 12 projects, all of them must be "consumed". Individual works will **not** be considered.
- Each project contains an exactly specified input device and an output device that **must be** incorporated in the project. Parameters of the devices are to be learned from the provided datasheets. The design and specification of the circuit **must conform** to the specified devices in terms of current, voltage and power rating, device impedance, special pinout, etc.
- All circuits has the following structure: (1) input signal conditioning stage (e.g. divider section), (2) signal conversion stage based on one or more operational amplifiers (configured as comparator, level shifter, etc.) and (3) output driving stage. This latter can be the output of an op-amp, but note that the output of an op-amp cannot drive high current devices (such as a motor, or power LED) directly. In this case, an external transistor may be necessary. Always check the ratings of the op-amp. Note that some of the tasks could be more or less realized using power supplies and some resistors, but this is not the point: what is expected to build your circuit around one or more op-amps that will give much more flexible design and accurate parameters.
- It is required that the power supply requirements (e.g. single or dual supply, or voltage requirement) of the circuit be mentioned and the necessary minimum (steady state) and maximum (on-state) currents be estimated based on the circuit design and datasheets. You do not need to design the supplies themselves, but they must appear in the circuit as connection pins/nodes. If, however, any reference voltage is required for the circuit, its (simple) design must be presented as part of the circuit.
- Try to realize the required specifications as closely as possible, but do not stick if you experience difficulties: the required circuit characteristics may not be necessarily linear.
- The project is complete once a report with the content specified in a template file is submitted. The requirements of each task can be found as entry text under each task title in the template file. Ideally, you should present a **6-page report with each section having a separate page**.
- **Advices:** All projects can be implemented with the knowledge we gained during lectures supplemented by reading Book Chapters 1 (DC), 2 (AC), 3 (diode), 4 (BJT) and 6 (op-amp) and the related Supplementary Material using basic laws (Ohm, KVL, KCL, equal circuits, etc.) and definitions by selecting appropriate resistors, capacitors, diodes and op-amps. BJTs are only required for high current drives.
- **Restrictions:** Advices section also refers to the requirement that the project must be implemented with methods gained during the course. Nevertheless, earlier knowledge and/or experience are always welcome if the project aim is not forgotten. Therefore, it is **strictly not allowed** to use any kind of microcontrollers, digital circuits, or complex-function integrated circuits containing more functionality than op-amps or transistors.